



PLACEMENT IS STILL A HUMAN GUESS

Every guess costs a full-wave run, and the run is the slowest thing in the building.

≈ 1 h

one full-wave run of a whole device

15 mm

keep-out at 2.4 GHz, or it fails RF test

dozens

iterations before sign-off

AND IT REPEATS,
EVERY DEVICE,
EVERY BAND

1.24B

smartphones a year, 6+ antennas
each

18.8→40B

IoT devices, 2024 to 2030

4→8

antennas per car by 2030

\$25–27B

antenna hardware, 2025

\$3.9–7.7B

the software and services layer on
top

ONE LOOP

The agent proposes, the solver scores, and it decides when to stop.

01 INPUT .blend + prompt Any device's build file and one sentence of intent.	02 INGEST Extract & classify Per-part geometry, materials, ground reference.	03 SCREEN Rule out anchors Which positions hit real metal. No solver runs.	04 JUDGMENT Agent proposes Type, position, length — with a stated reason.	05 PHYSICS PyNEC scores S11, bandwidth, VSWR, efficiency, clearance.	06 OUTPUT report.md The recommendation, the ranking, and the limits.
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EVIDENCE, NOT VERDICTS — BACK TO THE AGENT UNTIL IT CONCLUDES

A solve has to cost milliseconds, or there is no loop.

1.9 ms B5 · 850 MHZ	17 ms GPS L1	83 ms WI-FI / BT 2.4	404 ms N78 · 3.5 GHZ	2.0 s WI-FI 5 · 5.85 GHZ
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kevin

Apple iPhone 15 Pro

71.453 × 146.6 × 10.96 mm

DEVICE

Solver geometry

Apple iPhone 15 Pro

71.453 × 146.6 × 10.96 mm · 176 parts

Built-in catalogue spec.

Viewer

iPhone 15 Pro mesh

display only

191 drawn parts, none of them meshed. The solver reads the 176-part Apple iPhone 15 Pro spec above.

Load build file

A .blend is extracted on the backend and becomes solver geometry. A .glb is display only. Drop one on the card above or use the button.

TARGET BAND

per solve

B5

1.9 ms

GPS

17 ms

WiFi 2.4

83 ms

n78

404 ms

WiFi 5

2.0 s

Mesh pitch is $\lambda/10$ at the top of the band, so a solve costs about $f^{\wedge}3.7$ — 83 ms measured at 2.4835 GHz, 2.0 s at 5.85 GHz.

Wi-Fi / BT 2.4 GHz

2.400–2.483 GHz

S11 ≤ -8 dB Efficiency ≥ 50%

Clearance ≥ 4 mm

Radiator 31 × 1.8 × 0.9 mm

Type IFA / monopole / ceramic_chip

No solve another band

none

Iso Front Back Edge Top

Antennas

Keep-out

Labels

WiFi 2.4 -15 dB

Explode 0%

Chosen All for band

1 of 1 bands met

each track spans its own band · tick = simulated resonance

WiFi 2.4

-14.7 dB

Candidates (29) Report Evidence

BAND	PLACEMENT	S11 DB	F _c GHZ	H	VERDICT
WiFi...	top 24,128...	-14.7	2.499	97%	Pass
	top 24,128...	-14.6	2.499	97%	Pass
	top 24,128...	-20.4	2.327	39%	Fail
	top 24,128...	-10.8	2.388	77%	Fail

Wi-Fi / BT 2.4 GHz

2.400–2.483 GHz

Fail

IFA 32.1 mm at top 24,128 mm (escape 50%) · 28 other candidates in ...

S11 (dB)

target -8 dB

115 MHz at -6 dB

Frequency (GHz)

F _c GHZ	S11 DB	BW MHZ	H	VSWR	GAIN DBI
2.480	-15.1	115	96%	1.52	0.0

AGENT

Finished 0:01

SPEC

Mock Edit spec

Where should the antennas be placed in this phone? Pick the type, target band and expected performance for each, and...

c002_ifa_p10__length_mm=32.5 (IFA, Wi-Fi / ..

FAIL

S11 -15.8 dB at 2.442 GHz · BW 134 MHz · eff 85% · VSWR 2.29

PEC wire-grid model; worst in-band S11 -4.8 dB

c002_ifa_p10__length_mm=32.6 (IFA, Wi-Fi / ..

FAIL

S11 -16.7 dB at 2.442 GHz · BW 115 MHz · eff 80% · VSWR 2.59

PEC wire-grid model; worst in-band S11 -4.2 dB

c002_ifa_p10__length_mm=32.7 (IFA, Wi-Fi / ..

FAIL

S11 -16.2 dB at 2.423 GHz · BW 134 MHz · eff 76% · VSWR 2.93

PEC wire-grid model; worst in-band S11 -3.8 dB

Iteration 3 scored: trend converging, best so far

c002_ifa_p10__length_mm=31.9.

Requirements met — concluding.

Accepted

Top: c002_ifa_p10__length_mm=31.9 score 0.747 (all requirements pass); trend converging.

Recommendation

IFA at 23.5, 127.5, 9.6 mm for Wi-Fi / BT 2.4 GHz, length 31.9 mm — S11 -14.7 dB, efficiency 97%, every requirement met. Top: c002_ifa_p10__length_mm=31.9 score 0.747 (all requirements pass); trend converging.

Report ready (report.md, run.json, S11 CSVs).

Solves

29/29

LIVE CAPTURE · ONE STUDY, START TO REPORT, ON THIS MACHINE

IF THE BACKEND IS DOWN THE UI SAYS SO AND DRAWS NOTHING, RATHER THAN A STAND-IN

KEVIN

IT RUNS

Nothing on screen is a mock-up of a result.

/ 29 solves, one second.

Every candidate on the list has a real response behind it.

/ Every verdict has a reason.

Pass or fail, with the margin and the requirement that decided it.

/ The agent’s reasoning is the transcript.

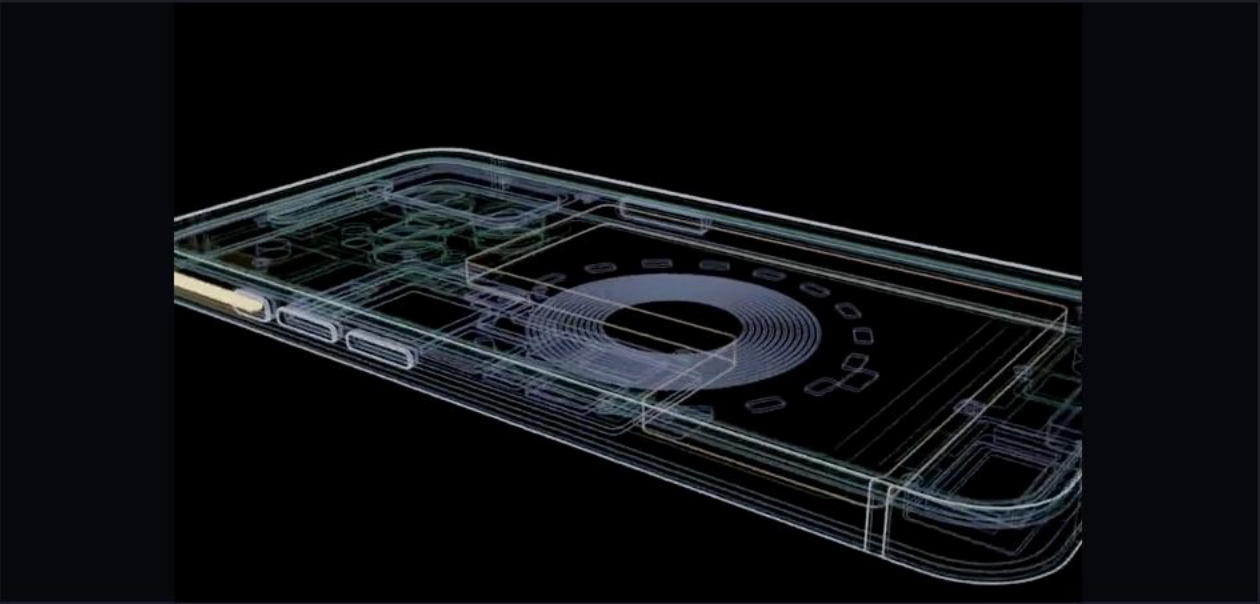
Not a progress bar — the actual argument it made, candidate by candidate.

The device is an input, not an assumption: drop in another build file and the same loop runs against it.

PASS FAIL PYNEC THREE.JS

SCREENING

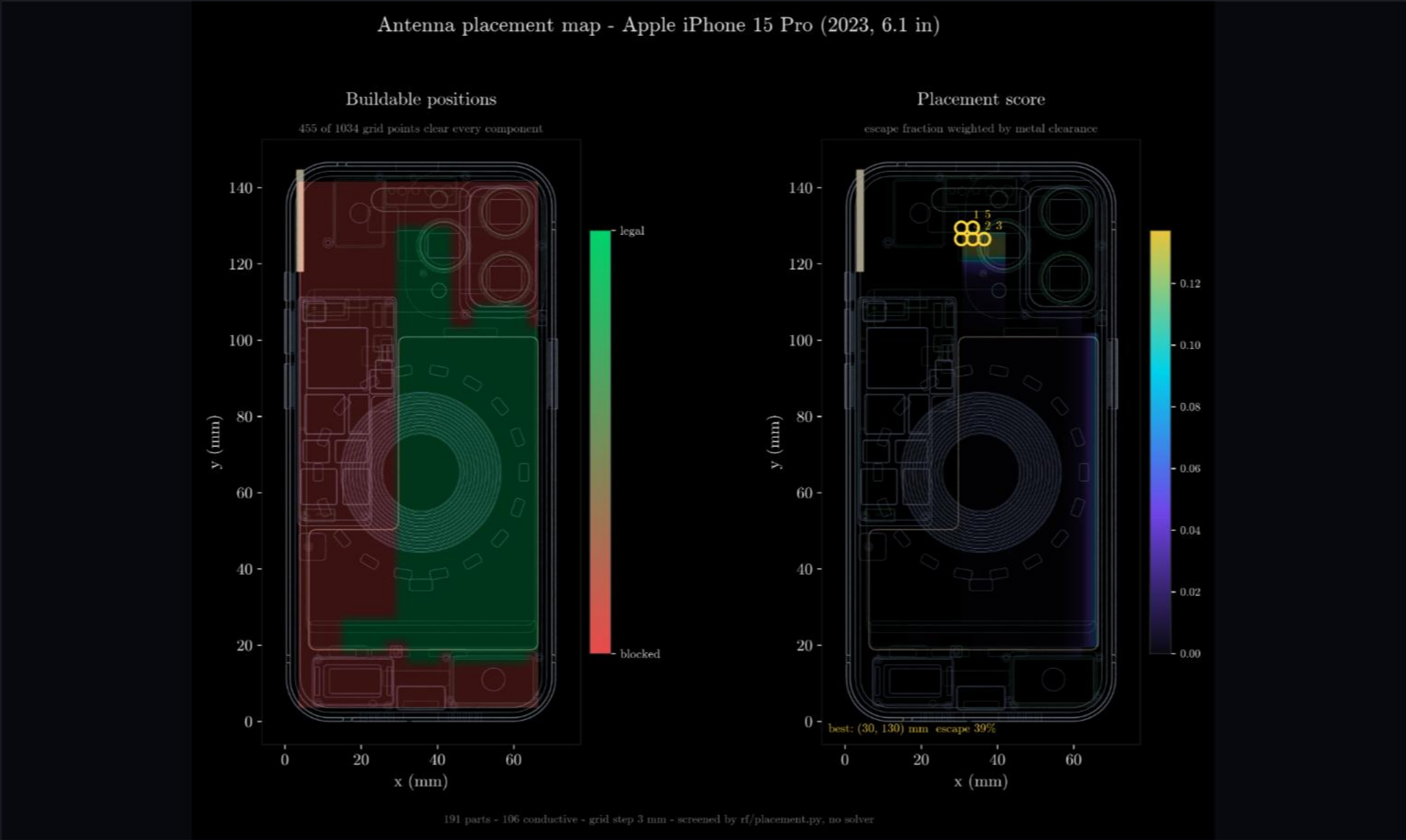
Before it proposes anything, it works out where an antenna physically cannot go.



THE GEOMETRY IT READS – 176 PARTS, RENDERED FROM THE BUILD FILE

WHAT THE SCREENING PASS RETURNS

Parts read	191 • 106 conductive
Grid points tested	1 034 @ 3 mm
Clear of every component	455
Anchors legal	5 of 20

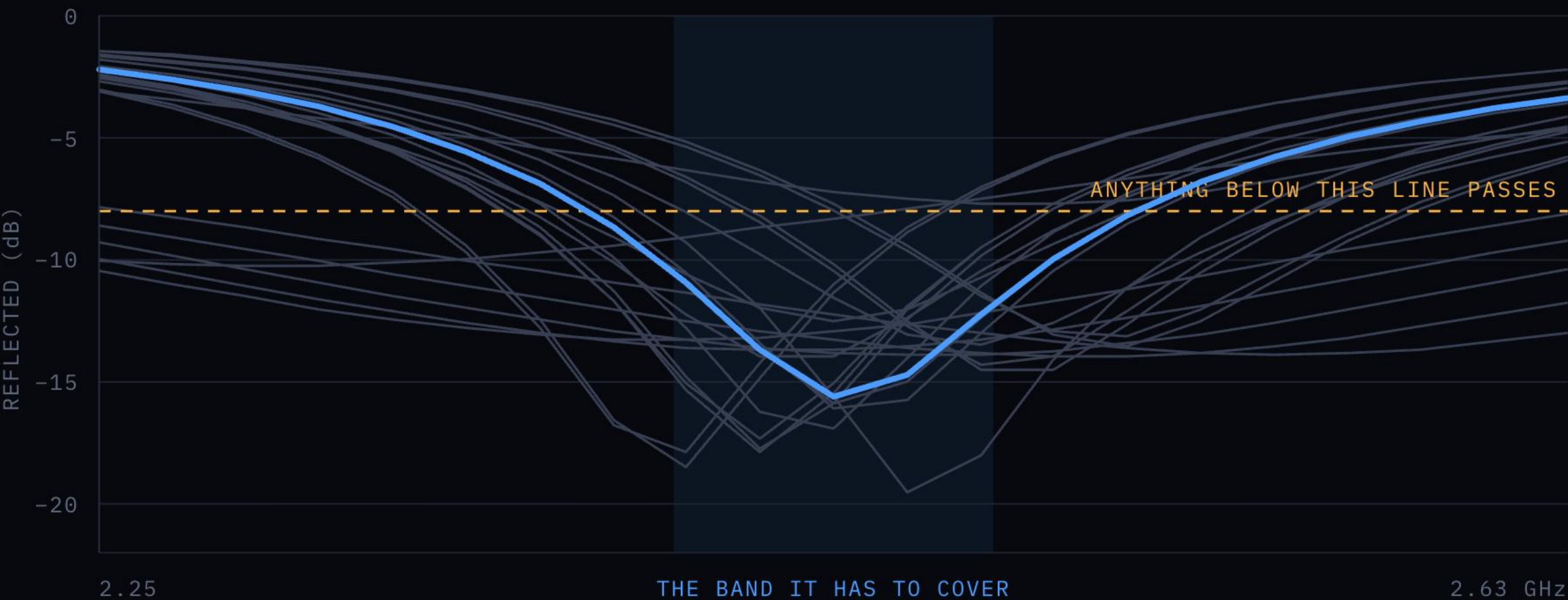


LEFT: EVERY POSITION THAT CLEARS ALL 191 PARTS. RIGHT: THE SAME VOLUME SCORED BY HOW MUCH OF THE SPHERE THE SIGNAL CAN LEAVE THROUGH – THE FIVE SURVIVING ANCHORS, NUMBERED.

RUN run_local · WI-FI / BT 2.4 GHZ

EVERY REQUIREMENT MET

It tried 41 positions. It kept the one it could defend.



ONE FAINT LINE PER SIMULATED CANDIDATE, IN THE ORDER THE LOOP PRODUCED THEM · THE ACCEPTED DESIGN IN BLUE · DRAWN FROM THE RUN'S OWN CSVs

IFA AT ANCHOR E_R2, ARM 30.75 MM · WORST IN-BAND -10.94 DB AGAINST A -8 DB SPEC · PEC WIRE-GRID MOM: DIRECTIONAL, NOT CERTIFICATION-GRADE — OPENEMS CONFIRMS THE WINNER

KEVIN

97%

of the power fed in leaves as radio, instead of bouncing back down the cable.

2×

the bandwidth the band actually needs — 172.6 MHz of headroom over an 83.5 MHz channel.

3

rounds of proposing and simulating before the agent called it converged.

ECONOMICS

The compute was never the expensive part. The calendar attached to it was.

≈40,000×

evaluations per unit of solver time – which is the whole reason an agent can sweep candidates instead of picking one and hoping.

MEASURED · 41 SOLVES IN 3.7 S

Fidelity is not equivalent, and that is the point. The in-loop model **selects**; openEMS **confirms** the winner. What collapses is the cost of asking the question, not the cost of the final answer.

ONE 41-CANDIDATE PLACEMENT STUDY	FULL-WAVE PATH	WITH KEVIN
Solver time	≈ 41 h	3.7 s
Wall clock to a ranked answer	≈ 1 working week	under a minute
Engineer time attached to it	≈ 10 h	< 1 h
Engineer cost of the study	\$640 – \$1,080	≈ \$60 – \$110

≈ 90%

off the engineer cost of a placement sweep – modelled

83 ms

per Wi-Fi 2.4 GHz solve, measured on this device

\$0

of solver licence – PyNEC and openEMS are both open source

Arithmetic: 41 × 91 ms = 3.7 s against 41 × 1 h. Engineer time at 15 min of setup and review per configuration; fully-loaded RF engineer \$130K-\$220K/yr → \$63-\$106 per hour.

BUILT AT THE HACKATHON

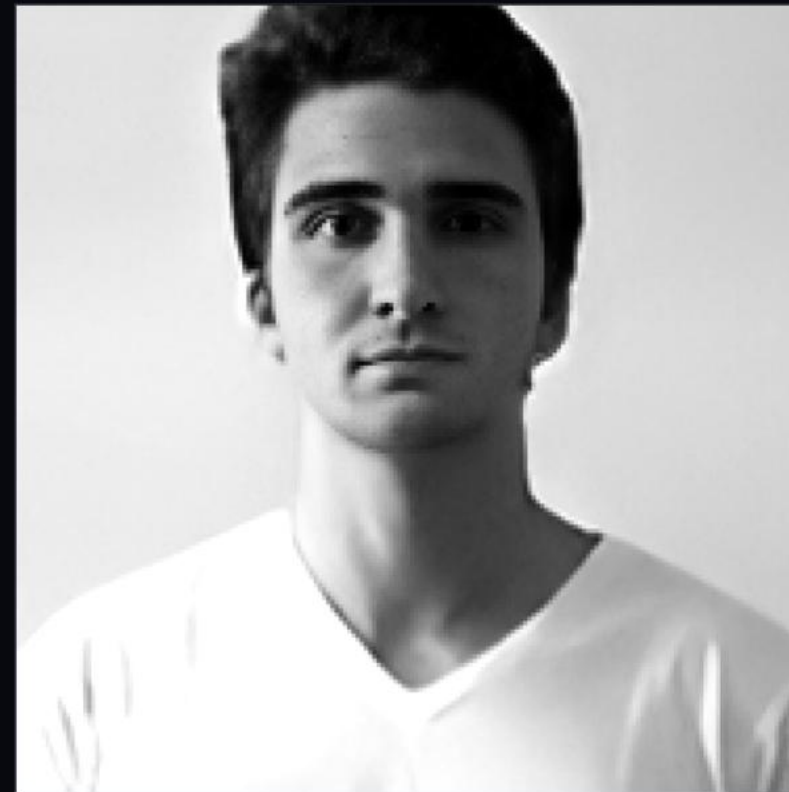
Four people, four seams.



**Aleksandr
Gorbunov**

Agent & systems integration

Devin API · orchestrator



**Damià
Vicens Ramis**

3D modelling & simulation

linkedin.com/in/dvr168



**Amr
Moussa**

Visualisation & UI

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**Rauf
Boudenne**

RF research & simulation

openEMS · PyNEC calibration

kevin knows where.

Build file in. Defensible placement out.